Phytopathology® • 2021 • 111:1080-1090 • https://doi.org/10.1094/PHYTO-08-20-0322-IA

Phytopathology®
REVIEW

Versatile Applications of the CRISPR/Cas Toolkit in Plant Pathology and Disease Management

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Accepted for publication 20 December 2020.

ABSTRACT

New tools and advanced technologies have played key roles in facilitating basic research in plant pathology and practical approaches for disease management and crop health. Recently, the CRISPR/Cas (clustered regularly interspersed short palindromic repeats/CRISPR-associated) system has emerged as a powerful and versatile tool for genome editing and other molecular applications. This review aims to introduce and highlight the CRISPR/Cas toolkit and its current and future impact on plant pathology and disease management. We will cover the rapidly expanding horizon of various CRISPR/Cas applications in the basic study of plant–pathogen interactions, genome engineering of plant disease resistance, and molecular diagnosis of diverse pathogens. Using the citrus greening disease as an example, various CRISPR/Cas-enabled strategies are presented to precisely edit the host genome for disease resistance, to rapidly detect the pathogen for disease management, and to potentially use gene drive for insect population control. At the cutting edge of nucleic acid manipulation and detection, the CRISPR/Cas toolkit will accelerate plant breeding and reshape crop production and disease management as we face the challenges of 21st century agriculture.

Keywords: biotechnology, CRISPR/Cas, disease control, disease resistance, genetics, genome editing, host parasite interactions, host resistance, molecular, pathogen detection

The ever-evolving arms race between plants and their pathogens often leads to devastating plant diseases, which pose major challenges for modern agriculture. During the past decades, a diverse array of biological and agricultural technologies has been developed to facilitate the elucidation of host–pathogen interactions, the breeding of disease resistant varieties, and timely detection and accurate diagnosis of plant pathogens for disease management. However, new and advanced technologies and tools are still much needed to improve plant health and crop production in the midst of climate change and the increasing global population. Recently, the invention and applications of CRISPR/Cas (clustered regularly interspersed short palindromic repeats/CRISPR associated) technology have revolutionized the field of genome editing and greatly contributed to the rapid advancements in biotechnology, biomedicine and agriculture (Adli 2018; Gao 2018). Increasing examples have shown that the CRISPR/Cas technology has the potential to transform plant pathology research and disease management by providing a series of molecular toolkits to edit, modify, or detect plant and pathogen DNAs and RNAs at an unprecedented pace and precision (Razzaq et al. 2019; Shelake et al. 2019; Zhao et al. 2020). This review will introduce the CRISPR/Cas technology and its applications in plant pathology and disease management, particularly in the areas of the host–pathogen interaction, plant disease resistance, and pathogen diagnostics.

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Funding: This work was supported in part by the U.S. Department of Agriculture National Institute of Food and Agriculture grant 2020-67013-31546 and Hatch Appropriations under Project PEN04659 and Accession 1016432 to Y. Yang.

The author(s) declare no conflict of interest.

CRISPR/CAS: A REVOLUTIONARY MOLECULAR TOOLKIT FOR GENOME ENGINEERING

The CRISPR/Cas system is a form of adaptive immunity in archaea and bacteria against virus and other foreign genetic elements through targeted interference or cleavage of nucleic acids (Barrangou et al. 2007; Brouns et al. 2008; Marraffini and Sontheimer 2008). The RNA-guided nucleases of the CRISPR/Cas system are capable of binding to and cleaving specific nucleic acid sequences upon viral infection to provide antiviral interference and bacterial immunity (Barrangou et al. 2007). Through a process called adaptation, bacteria capture segments of foreign genetic elements and incorporate them into their genomic DNAs at the

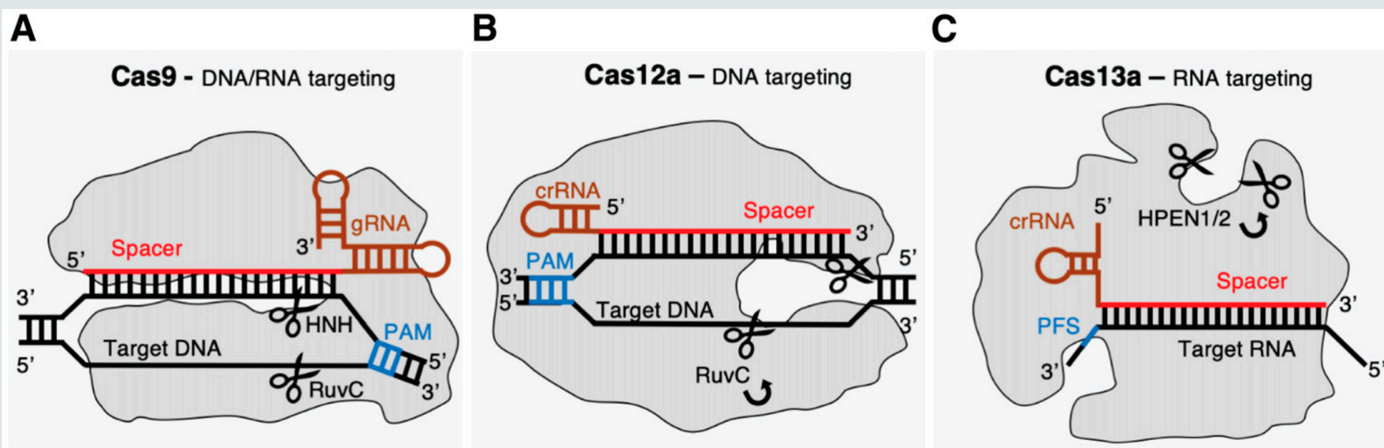
CRISPR array locus (Sternberg et al. 2016). Transcription and processing of the CRISPR array results in the generation of CRISPR RNAs (crRNAs) that help guide Cas nucleases for targeted and specific cleavage of nucleic acid sequences through complementary base-pairing (Brouns et al. 2008; Marraffini and Sontheimer 2008). The class 2 CRISPR systems, commonly employed for genome editing, are comprised of single, RNA-guided Cas nucleases that mediate targeted interference or cleavage of nucleic acids (Koonin et al. 2017).

Cas9 is a class 2 type II DNA-targeting endonuclease and the first Cas effector to be utilized for genome editing (Cong et al. 2013; Jinek et al. 2012) (Box 1). The bacterial crRNA (CRISPR RNA) and tracrRNA (trans activating crRNA) work together to specifically guide Cas9 to the target sequence for DNA cleavage. These two small RNA molecules can be artificially combined into a single guide RNA (sgRNA or gRNA) to create a simple, two-component system consisting of Cas9 and gRNA for in vitro and in vivo genome editing (Cong et al. 2013; Jinek et al. 2012). Once the Cas9/gRNA complex has stably bound to the target DNA sequence adjacent to a protospacer adjacent motif (PAM), Cas9 will unwind and cut the DNA to generate a double-strand break (DSB). With a replacement of 20 nucleotides in the spacer region, a gRNA can redirect Cas9 to a different target for DSB formation. Due to its simplicity, programmability and multiplex editing capability in comparison with the previous genome editing tools such as zinc finger nuclease (ZFN) and transcription activator-like effector nuclease (TALEN), CRISPR/Cas9 has rapidly become the premier choice for diverse genome editing applications in biotechnology, medicine and agriculture.

Currently, *Streptococcus pyogenes* Cas9 (SpCas9) is the most commonly used Cas nuclease for genome editing (Box 1). However, other bacterial Cas9 orthologs and artificially engineered Cas9 variants have been developed to provide enhanced specificity (i.e., high fidelity) and broader targetability via PAM variation. Aside from Cas9, other class 2 Cas nucleases such as Cas12a (Cpf1) (Zetsche et al. 2015) and Cas13a (Abudayyeh et al. 2016) have become increasingly popular for DNA or RNA editing and detection. These additional Cas

BOX 1

INTRODUCTION TO CAS NUCLEASES



Commonly used Cas nucleases Cas9, Cas12a, and Cas13a.

A, Cas9 is shown with the spacer (red) of gRNA (brown) bound to target dsDNA (black) proximal to protospacer adjacent motif (PAM) (blue). Cas9 is a class 2 type II nuclease that is able to bind and cut double-stranded (ds)DNA. Cas9 is directed toward its genomic target via crRNA and tracrRNA or a chimeric gRNA (fused crRNA with tracrRNA) (Jinek et al. 2012). Upon recognition of the PAM and complementary base-pairing of gRNA spacer with target DNA, Cas9 unwinds target DNA (Brouns et al. 2008), and its RuvC and HNH nuclease domains are activated to cleave the target and nontarget strands (scissors), generating a DNA double-strand break (Jinek et al. 2012).

B, Cas12a is shown with the spacer (red) of crRNA (brown) bound to target dsDNA (black) proximal to PAM (blue). Cas12a is a class 2 type V nuclease that is capable of cleaving single-stranded (ss)DNA and dsDNA (Zetsche et al. 2015; Chen et al. 2018). Following the binding to target DNA sequence adjacent to a PAM and complementary to the crRNA spacer, Cas12a nuclease will create a staggered DSB in the target and nontarget strands, generating an overhang on the nontarget stand (Zetsche et al. 2015). Upon activation of the RuvC nuclease domain, Cas12a has general ssDNase activity and is capable of collateral cleavage of ssDNA substrate (Chen et al. 2018).

C, Cas13a is shown with the spacer (red) of crRNA (brown) bound to RNA target sequence through complementary base-pairing. Cas13a is a class 2 type VI nuclease that targets ssRNA through the guidance of a single crRNA (Abudayyeh et al. 2016). Like the other Cas nucleases, Cas13a binds to its target sequence complementary to the crRNA spacer. Upon correct base-pairing and activation of Cas13a, the HPEN nuclease domain becomes active with general ssRNase activity (Abudayyeh et al. 2016; Gootenberg et al. 2017). Cas13a does not require a typical PAM in the target sequence, but a protospacer flanking sequence (PFS) is involved to allow greater targeting capabilities.

nucleases have further helped to expand the CRISPR/Cas toolkit and provide much increased versatility for genome engineering, disease diagnostics and other molecular applications beyond conventional genome editing (Murugan et al. 2017).

GENOME EDITING AND THE CRISPR/CAS TOOLKIT

A diverse set of CRISPR/Cas tools have been developed for genome editing based upon the flexible programmability and targetability of different types of Cas nucleases. The typical and most commonly used genome editing application is the generation of loss-of-function mutants. This process involves a sequence-specific nuclease such as Cas9 to generate a DSB at the target DNA site and relies on the endogenous cellular DNA repair pathway, nonhomologous end joining (NHEJ) to ligate the broken ends (Fig. 1A). In higher eukaryotes, including plants and animals, NHEJ is the prevalent pathway of DNA repair and frequently generates small insertions or deletions (indels) at the DSB site, causing frame-shift mutations in protein-coding sequences and, consequently, nonfunctional variants. In contrast, homology directed repair (HDR) uses a DNA donor template with homologous sequences, allowing exchange of the damaged sequences and more precise repair of DSBs at the target DNA site (Pawelczak et al. 2018). While HDR is a predominant form of DNA repair in bacteria and many fungi (Ayora et al. 2011), higher organisms have a low frequency of HDR outcomes. Due to the importance of precise editing in many biomedical and agricultural applications, major efforts are being made to overcome low HDR by manipulating DNA repair pathways in favor of HDR (Liu et al. 2019; Pawelczak et al. 2018), or taking alternative approaches such as prime editing (Lin et al. 2020).

Besides native Cas9 nuclease, single strand-cleaving nickase and deactivated Cas9 (dCas9) have been increasingly utilized to enable Cas platforms for single base editing, transcriptional regulation, epigenome editing and nucleic acid imaging. Base editing

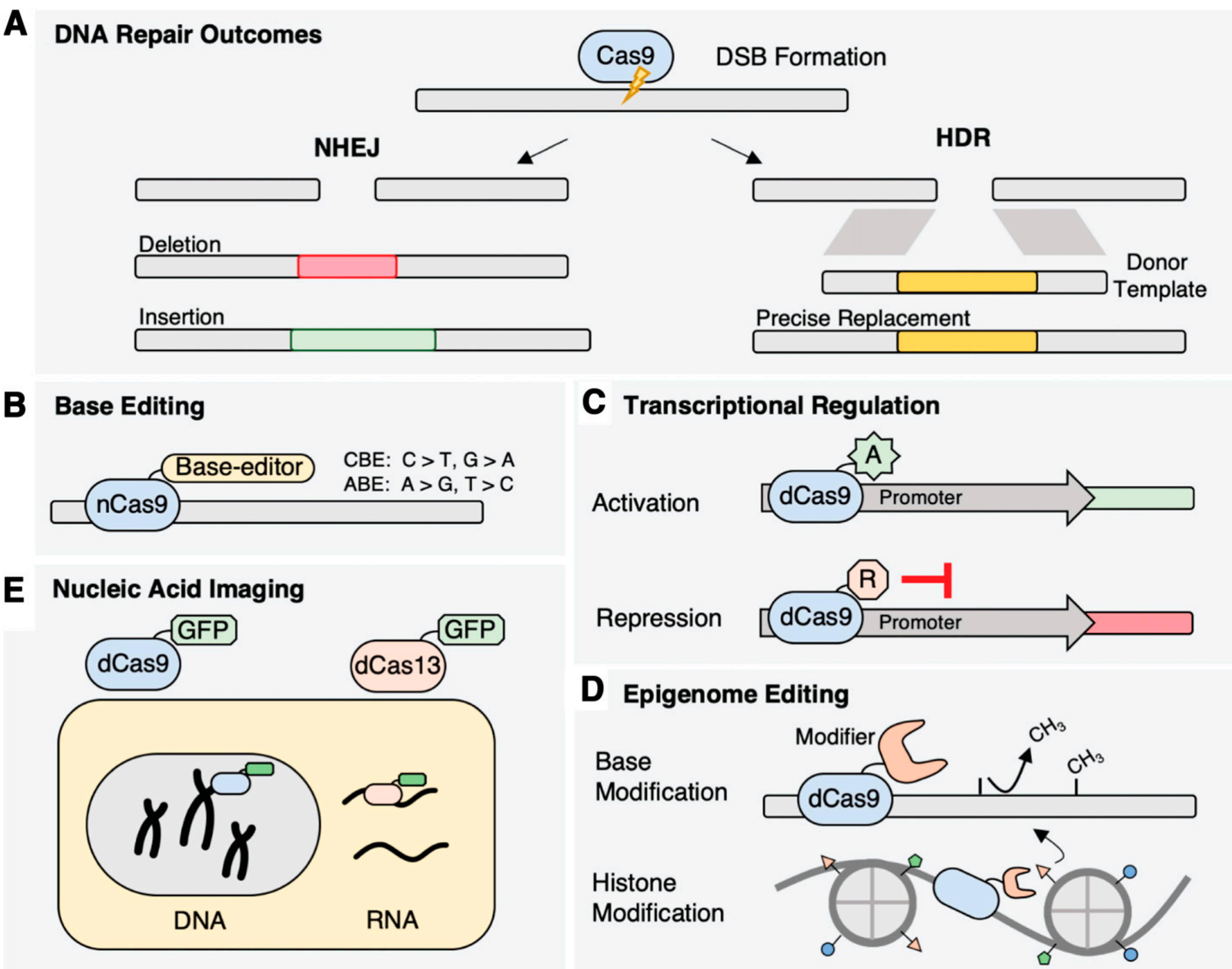


FIGURE 1

DNA repair and CRISPR/Cas-enabled genome editing. **A,** Typical cellular DNA repair pathways after Cas9 cleavage and generation of double strand break (DSB). NHEJ repair involves imprecise ligation of dsDNA ends which can result in the formation of indels, whereas HDR requires a donor DNA template containing flanking homologous sequence for precise gene replacement. **B,** Cytosine or adenine base editors for precise base editing without the generation of DSB. **C,** Deactivated Cas9 (dCas9) fused to transcriptional activator or repressor bound to a promoter region to up or down regulate gene expression. **D,** Epigenome editing with dCas9 fused with DNA methylase, demethylase or histone deacetylase. **E,** Nucleic acid imaging with dCas9 fused to fluorescent proteins such as GFP for fluorescent labeling of genomic DNA or with dCas13 fused to GFP for RNA labeling.

is a relatively new CRISPR/Cas tool that utilizes nickase or deactivated variants of Cas proteins fused to a cytidine or adenine deaminase (Fig. 1B) (Molla and Yang 2019; Rees and Liu 2018). As a result, it does not involve DSBs and requires no donor template. Single base conversion has been achieved with Cas9- or Cas12a-dependent base editors on DNA (Komor et al. 2016) or with Cas13a-dependent base editors on RNA (Cox et al. 2017). All four transition mutations (C > T or G > A, and A > G or T > C) have been achieved using cytosine base editor (CBE) and adenine base editor (ABE). Most recently, new base editors have also enabled C > G and C > A transversion mutations (Molla et al. 2020a). In plants, single base editing has been used as an alternative to HDR-mediated methods for editing single nucleotide polymorphisms (SNPs) corresponding to various agronomic traits (Li et al. 2018a, 2020; Molla et al. 2020b; Shimatani et al. 2017; Zong et al. 2017).

In addition to altering nucleic acid sequences, CRISPR/Cas toolkits have been created to up or down regulate the expression of target genes. Altered regulation is typically done through direct fusion of dCas9 with a transcriptional activator/repressor or recruitment of transcriptional regulators by an RNA aptamer fused with the gRNA or an epitope tag (Fig. 1C). By direct fusion or recruitment of transcriptional activator VP64 to dCas9, target gene activation or repression was achieved in diverse chromatin contexts in *Arabidopsis* (Lowder et al. 2018; Papikian et al. 2019). Furthermore, gene expression could be modulated via DNA or histone epigenome editing, which involves the fusion or recruitment of methyltransferases, demethylase, or histone deacetylase to dCas9 (Fig. 1D) (Gallego-Bartolomé et al. 2018; Papikian et al. 2019). Another popular application of deactivated Cas proteins is sequence-specific nucleic acid imaging for cell biology studies (Wu et al. 2019). Imaging has been achieved with dCas9 fusion to fluorescent proteins and the recruitment of organic based dyes, or nanoparticles, to study single or multiple genomic loci in living cells (Fig. 1E). These tools have been widely utilized for investigating chromatin architectures and transcript localization in healthy and diseased states of mammalian cells to gain insights into molecular and cellular processes associated with normal physiology and pathogenesis. Currently, CRISPR/Cas-based imaging, while in its infancy in plant systems, has been reported in *Nicotiana benthamiana* to study telomere movements during cell division (Dreissig et al. 2017).

With the development of CRISPR/Cas tools, a concern has arisen with regards to off-targeting effects associated with genome editing, especially when considering their potential impact on biomedical and therapeutic treatments. High-fidelity or enhanced Cas nucleases have been engineered to improve the editing specificity (Kleinstiver et al. 2016; Slaymaker et al. 2016). In addition, in vitro and in vivo assays as well as in silico bioinformatics tools have been developed for genome-wide off-target analyses and prediction of specific protospacers for gRNA design (Razzaq et al. 2019). For example, bioinformatics tools and websites such as CRISPR-PLANT (Minkenberg et al. 2019; Xie et al. 2013), CRISPR-P (Lei et al. 2014; Liu et al. 2017) and EuPaGDT (eukaryotic pathogen CRISPR gRNA/DNA design tool) (Peng and Tarleton 2015) were established to allow the prediction and design of specific gRNA spacers for SpCas9 and other Cas nucleases. Recent studies indicate that there is a minimal risk of genome-wide off-targeting in plants when using Cas9 with a highly specific gRNA spacer, since off-target editing was detected at a lower level than natural variation (Young et al. 2019).

CRISPR/CAS APPLICATIONS IN BASIC RESEARCH OF PLANT–PATHOGEN INTERACTIONS

The CRISPR/Cas toolkits have been mostly utilized for targeted mutagenesis in hosts and pathogens to facilitate the functional analysis of plant and microbial genes and the mechanistic studies of host immunity and pathogen virulence (Fig. 2A). Genome editing and targeted mutagenesis have been achieved using CRISPR/Cas in model plants as well as agriculturally important cereals and specialty crops (Shan et al. 2013; Wang et al. 2019; Xie and Yang 2013). There are also increasing number of reports on CRISPR/Cas-mediated genome editing and target mutagenesis in fungi and oomycetes (Schuster and Kahnmann 2019). Plant pathogens, including *Aspergillus* spp. (Fuller et al. 2015; Nødvig et al. 2015, 2018), *Alternaria alternata* (Wenderoth et al. 2017), *Ashbya gossypii* (Jiménez et al. 2020), *Ustilaginoidea virens* (Liang et al. 2018), *Ustilago maydis* (Schuster et al. 2016, 2018), *Fusarium fujikuroi* (Shi et al. 2019), *F. graminearum* (Gardiner and Kazan 2018), *F. oxysporum* (Wang et al. 2018), *M. oryzae* (Arazoe et al. 2015; Foster et al. 2018) *Leptosphaeria maculans* (Idnurm et al. 2017), *Phytophthora capsici* (Miao et al. 2018), *P. sojae* (Fang and Tyler 2016), *P. palmivora* (Wang et al. 2019), *Sclerotinia sclerotiorum* (Li et al. 2016b), *Shiraia bambusicola* (Deng et al. 2017), *Sporisorium scitamineum* (Lu et al. 2017), *Ustilago tricophora* (Huck et al. 2019), and *Huntiella omanensis* (Wilson and Wingfield 2020), have been edited with CRISPR/Cas approaches. While higher eukaryotes, including plants, predominantly repair DSBs through the NHEJ pathway, many fungi tend to not follow the NHEJ repair process. It has been shown that some fungi rely heavily on HDR for DSB repair, while other fungi are capable of both NHEJ and HDR (Vyas et al. 2018).

The utility of CRISPR/Cas9 toolkits have been demonstrated for loss-of-function analysis and molecular characterization of the plant–microbe interactions during the host–pathogen recognition (e.g., PAMP or effector-triggered immunities), signal transduction, and subsequent host defense responses. For example, CRISPR/Cas9-mediated mutagenesis was used to characterize NRG1, a CC-NB-LRR protein, and its role in plant immunity, leading to the discovery that NRG1 may act as a key conserved component in TIR-NLR-mediated signaling pathway (Qi et al. 2018). After the map-based cloning of a non-NLR rice resistance gene, *Ptr*, CRISPR/Cas9-enabled loss-of-function analysis was also used to verify its role in mediating broad-spectrum resistance against the rice blast fungus (*Magnaporthe oryzae*) (Zhao et al. 2018). In addition, the characterization of defense-related WRKY transcription factors in *Brassica napus* benefited from the use of CRISPR/Cas9-mediated targeted mutagenesis (Sun et al. 2018). While targeted mutation of *BnWRKY11* did not alter host resistance, knockout of *BnWRKY70* increased host resistance to *Sclerotinia sclerotiorum*. With their increasing application in plants, CRISPR/Cas tools will become ubiquitously used and provide a versatile platform for investigating the plant defense responses.

The majority of CRISPR/Cas applications in microbes have been focused on metabolic engineering of industrially relevant fungi and bacteria to improve secondary metabolite production as well as the removal of toxic compounds (Shi et al. 2017). For example, CRISPR/Cas9 has been used to engineer strains of *Fusarium fujikuroi* to improve the production of gibberellic acid, which is a widely used plant growth regulator in agriculture such as fruit production (Shi et al. 2019). CRISPR/Cas9-mediated mutagenesis was also used to study the production of various antifungal compounds and volatile organic metabolites in *Bacillus* species to evaluate their plant-growth promoting effects on *Brassica rapa* (Yi et al. 2018). Recently, CRISPR/Cas9 tools have been used for targeted mutagenesis and functional characterization of effectors and other virulence genes in plant pathogens. CRISPR/Cas9 was used for disruption and replacement of the RXLR effector gene encoding Avr4/6 in *Phytophthora sojae*, facilitating the characterization of its corresponding resistance genes (*Rps4* and *Rps6*) in soybean (Fang and Tyler 2016). CRISPR/Cas9 has also been used as an efficient genome editing tool for a comparative analysis of secreted proteins in *Ustilago maydis* and other related smut basidiomycetes (Schuster et al. 2018). In addition to targeted mutagenesis, CRISPR/Cas9 has

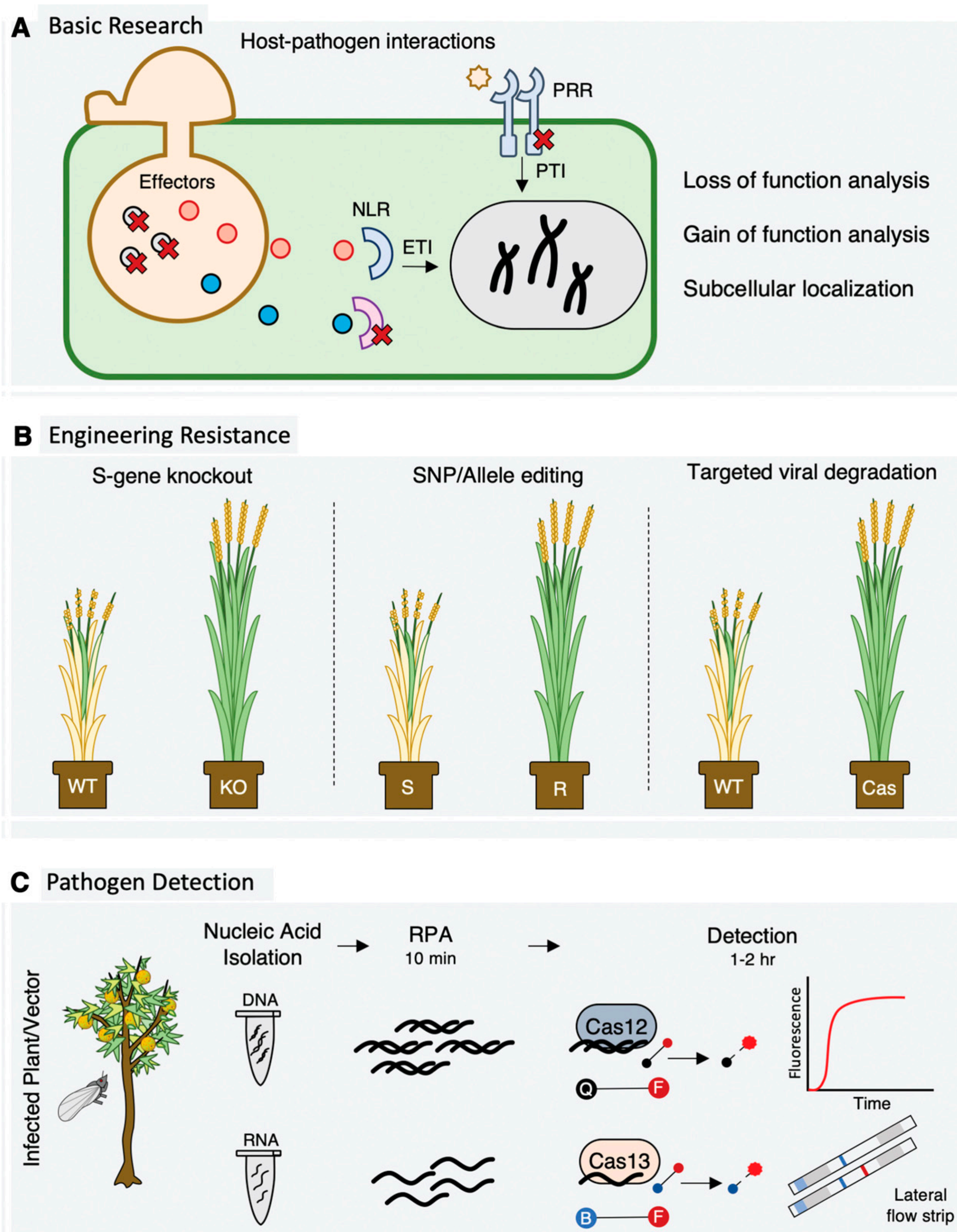


FIGURE 2

Major applications of CRISPR/Cas in plant pathology and disease diagnostics. **A,** Utilization of the CRISPR/Cas toolkit for basic research of host–pathogen interactions, which include targeted mutagenesis and loss of function analysis in both pathogen and host to understand the mechanisms of pathogen virulence and host resistance such as PAMP-triggered immunity (PTI) and effector-triggered immunity (ETI). **B,** Utilization of CRISPR/Cas for engineering plant disease resistance, which include targeted mutation of *S* genes or negative regulators of plant immunity, single nucleotide polymorphism (SNP)/allele editing in promoter or coding sequences to convert susceptible alleles to resistance alleles, and targeted degradation of viral DNA or RNA by expressing Cas nucleases in transgenic plants for virus control. **C,** Cas-based pathogen detection involves isolation of nucleic acids from plant disease samples, isothermal amplification of target nucleic acid through a recombinase polymerase amplification (RPA) reaction, and Cas nuclease detection. Upon recognition of the amplified target, Cas12a and Cas13a are activated with single-stranded (ss)DNase and ssRNase activity, respectively, to degrade fluorescent reporter for qualitative or quantitative detection of pathogen nucleic acids.

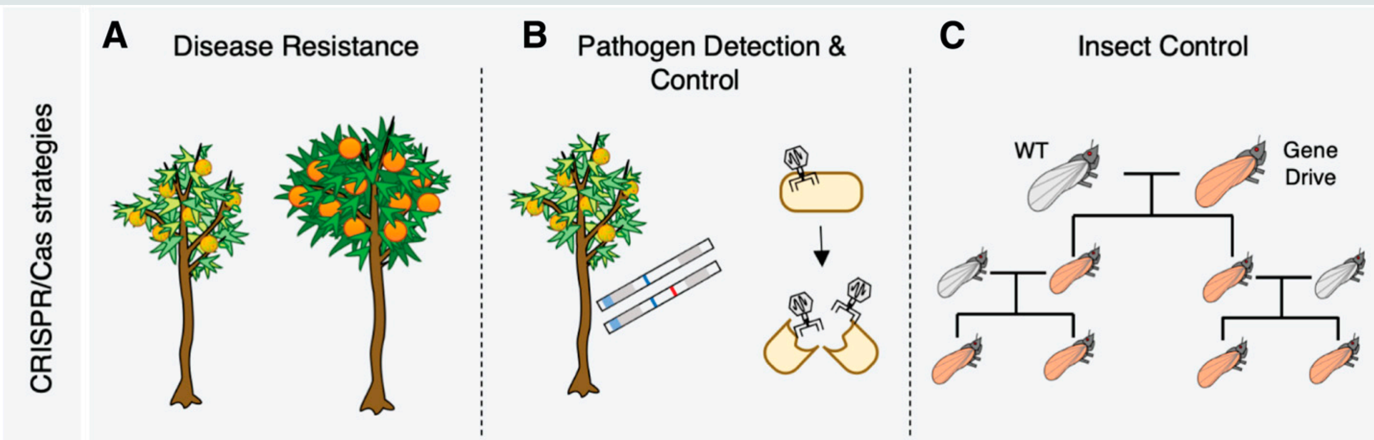
been utilized to tag effector proteins with fluorescent reporters such as GFP and RFP via HDR (Chen et al. 2019b). As a result, CRISPR/Cas toolkits will be increasingly used to aid in the cellular localization of protein effectors, functional discovery of pathogenicity genes, and mechanistic elucidation of pathogen virulence.

CRISPR/CAS APPLICATIONS IN GENOME ENGINEERING OF PLANT DISEASE RESISTANCE

CRISPR/Cas technologies have significant advantages for plant breeding and crop improvement, including engineering plant disease resistance (Chen et al. 2019a) (Fig. 2B, Box 2, Table 1). The most prominent utilization of CRISPR/Cas in engineering plant disease resistance has been achieved through targeted mutagenesis of susceptibility (S) genes or negative regulator of plant immunity (Zaidi et al. 2018). The mildew resistance locus O (*MLO*) is a conserved *S* gene originally identified from barley. Targeted mutation of the *MLO* homologs in wheat (Wang et al. 2014) and tomato (Nekrasov et al. 2017) was shown to provide enhanced disease resistance by disrupting host recognition and penetration of the powdery mildew fungal pathogens. In *Arabidopsis*, EDR1 acts a negative regulator of plant defense signaling. Simultaneous mutation of three TaEDR1 homologs via CRISPR/Cas9 also led to powdery mildew resistance in wheat (Zhang et al. 2017). DMR6, which stands for downy mildew resistance, encodes another negative regulator of plant immunity that impacts salicylic acid homeostasis. CRISPR/Cas9-mediated mutation of tomato DMR6 ortholog rendered the mutant plants with broad-spectrum resistance against oomycete and bacterial plant pathogens (Thomazella et al. 2016). In addition, recessive resistance to potyviruses has been achieved by CRISPR/Cas9-enabled mutagenesis of the eukaryotic translation

BOX 2

CRISPR/CAS-BASED STRATEGIES FOR MANAGING CITRUS GREENING DISEASE



Manipulation or detection of host, pathogen, and insect with CRISPR/Cas.

A, Targeted mutagenesis of susceptibility genes and allele editing of *R* genes or defense genes with CRISPR/Cas may reduce citrus greening symptoms and enhance host resistance. **B,** Cas12a- and Cas13a-based sensitive detection of citrus greening pathogens in citrus plants and insect vectors will aid in the quarantine, exclusion, and eradication of infected materials and timely applications of pesticides. **C,** Lastly, CRISPR/Cas-based gene drive may be employed for controlling vector populations or rendering psyllids resistant to bacterial infection or incapable of disease transmission.

Citrus greening disease, also known as Huanglongbing (HLB), is a widespread and economically important disease of citrus. The disease is caused by a phloem-limiting, fastidious gram-negative bacterium *Candidatus* Liberibacter spp. with '*Candidatus* L. asiaticus' (CLas) exerting the most extensive damage in many citrus growing regions of the world (Wang et al. 2017). This pathogen is transmitted by grafting and sap-sucking psyllids *Diaphorina citri* and *Trioza eryteae*. After transmission, the bacterium accumulates in the phloem, restricting nutrient transport throughout the plant and resulting in chlorosis of leaves and shoots, blotchy necrotic spots, stunted shoot and root development, and small, green, malformed fruits, and eventually death of the plant.

CRISPR/Cas technologies can be used to detect the pathogen (CLas) and modify the host (citrus) or the insect (psyllid) for potential disease management. CRISPR/Cas9 has been used to edit susceptibility genes for enhanced resistance to citrus canker disease caused by *Xanthomonas axonopodis* pv. *citri* (Jia et al. 2017; Peng et al. 2017). Similarly, CRISPR/Cas-enabled genome editing can also be used to improve host resistance to citrus greening disease via targeted mutation of disease susceptibility genes or negative regulators (repressors) of plant immunity, precise knock-in of transcriptional enhancer to boost citrus immunity, and allele editing to convert disease susceptible alleles to resistant alleles. Besides editing the host genome, it is possible that CRISPR/Cas-based antibacterial agents may be delivered via bacteriophage or other approaches to target and eliminate CLas. Due to its high sensitivity and specificity, Cas nuclease-based diagnostics will likely be an excellent tool for early detection of low titer CLas in the asymptomatic tissues, which will facilitate disease management via prompt removal of infected trees. Since chemical-based management has been inadequate or insufficient for psyllid control due to implementation costs, evolving insecticide resistance, food safety, and environmental concerns, CRISPR/Cas-based gene drive may provide an alternative approach to manage insect vector population and mitigate pathogen spread in the future.

TABLE 1
CRISPR/Cas applications for improving disease resistance against plant pathogens

Plant	Modification	Target	Pathogen	Reference
Tomato	Gene knockout	*SlMlo1*	*Oidium neolycopersici*	(Nekrasov et al. 2017)
Tomato	Gene knockout	*DMR6*	*Pseudomonas syringae* *Phytophthora capsici* *Xanthomonas* spp.	(Thomazella et al. 2016)
Arabidopsis	Gene knockout	*eIF4E*	*Turnip mosaic virus*	(Pyott et al. 2016)
Cucumber	Gene knockout	*eIF4E*	*Cucumber vein yellowing virus* *Zucchini yellow mosaic virus* *Papaya ring spot mosaic virus*	(Chandrasekaran et al. 2016)
Cassava	Gene knockout	*eIF4E (nCBP-1, nCBP-2)*	*Cassava brown streak virus* *Ugandan cassava brown streak virus*	(Gomez et al. 2019)
Grape	Gene knockout	*WRKY52*	*Botrytis cinerea*	(Wang et al. 2018)
Cacao	Gene knockout	*NPR3*	*Phytophthora tropicalis*	(Fister et al. 2018)
Apple	Gene knockout	*DIPM-1, DIPM-2, DIPM-4*	*Erwinia amylovora*	(Malnoy et al. 2016)
Wheat	Gene knockout	*TaMLO-A1, TaMLO-B1, TaMLO-D1*	*Blumeria graminis* f. sp. *tritici*	(Shan et al. 2013; Wang et al. 2014)
Wheat	Gene knockout	*TaEDR1*	*Blumeria graminis* f. sp. *tritici*	(Zhang et al. 2017)
Wheat	Gene knockout	*TaNFXL1*	*Fusarium graminearum*	(Brauer et al. 2019)
Rice	Gene knockout	*OsMPK5*	*Magnaporthe grisea* *Burkholderia glumae*	(Xie and Yang 2013)
Rice	Gene knockout	*SEC3A*	*Magnaporthe oryzae*	(Ma et al. 2017)
Rice	Gene knockout	*ERF922*	*Magnaporthe oryzae*	(Wang et al. 2016)
Rice	Gene knockout	*eIF4G*	*Rice tungro spherical virus*	(Macovei et al. 2018)
Rice	Gene knockout	*SWEET13*	*Xanthomonas oryzae* pv. *oryzae*	(Zhou et al. 2015)
Citrus	Promoter editing	*CsLOB1*	*Xanthomonas citri* subsp. *citri*	(Jia et al. 2017; Peng et al. 2017)
Rice	Promoter editing	*OsSWEET11* *OsSWEET13* *OsSWEET14*	*Xanthomonas oryzae* pv. *oryzae*	(Oliva et al. 2019)
Rice	Base editing	*Pi-ta*	*Magnaporthe oryzae*	(Ren et al. 2018)
Nicotiana benthamiana	Viral interference	CP, Rep, IR	*Bean yellow dwarf virus*	(Ji et al. 2015)
Arabidopsis	Viral interference	CP, Rep, IR	*Bean yellow dwarf virus*	(Ji et al. 2015)
Nicotiana benthamiana	Viral interference	LIR, Rep/RepA	*Beet severe curly top virus*	(Baltes et al. 2015)
Nicotiana benthamiana	Viral interference	CP, Rep, IR	*Tomato yellow leaf curl virus* *Beet curly top virus* *Merremia mosaic virus* *Tomato yellow leaf curl virus*	(Ali et al. 2015)
Nicotiana benthamiana	Viral interference	CP Rep IR	*Cotton leaf curl Kokhran virus* *Merremia mosaic virus*	(Zaidi et al. 2016)
Nicotiana benthamiana	Viral interference	GFP1 GFP2 CP HC-Pro	*Turnip mosaic virus*	(Aman et al. 2018)
Nicotiana benthamiana	Viral interference	ORF1 ORF2 CO 3′UTR	*Cucumber mosaic virus* *Tobacco mosaic virus*	(Zhang et al. 2018)
Arabidopsis	Viral interference	ORF1 ORF2 CO 3′UTR	*Cucumber mosaic virus* *Tobacco mosaic virus*	(Zhang et al. 2018)
Cassava	Viral interference	CR, AC1, AC2, AV2	*African cassava mosaic virus*	(Mehta et al. 2019)

initiation factor 4E (eIF4E) gene family in cassava (Gomez et al. 2019), *Arabidopsis* (Pyott et al. 2016), and cucumber (Chandrasekaran et al. 2016). Besides targeting the protein-coding sequences, CRISPR/Cas9 genome editing of the transcription activator-like effector (TALE)-binding elements within the promoters of *CsLOB1* in citrus (Jia et al. 2017; Peng et al. 2017) and *OsSWEET11/OsSWEET13/OsSWEET14* (Oliva et al. 2019) in rice was shown to confer disease resistance against multiple races of *Xanthomonas* pathogens. Targeted mutagenesis has also been utilized for conferring resistance to *F. graminearum* via CRISPR/Cas genome editing of mycotoxin-induced (deoxynivalenol) wheat transcription factor TaNFLX1 limiting the growth and establishment of the pathogen postinfection (Brauer et al. 2019).

Allele variations, including SNPs, are responsible for various agronomic traits including herbicide tolerance or disease resistance. In comparison with the HDR-based approach, single base editing is a more efficient strategy for precise base modifications in plants. Currently, base editing tools have been used to develop herbicide resistance (Li et al. 2018a; Shimatani et al. 2017; Zong et al. 2017), improved nitrogen efficiency (Lu and Zhu 2017) and disease resistance (Ren et al. 2018) in crops such as rice, wheat, or tomato. In addition to engineering *R* gene alleles, a library of novel alleles may be created by targeted mutagenesis or base editing in planta and screened for specific or broad-spectrum disease resistance (Li et al. 2020). Furthermore, CRISPR/Cas mutagenesis in the promoter regions of agronomically important genes creates a pool of potential dosage effect alleles and a continuum of quantitative trait variations (Rodríguez-Leal et al. 2017). Several groups used CRISPR/Cas genome editing to add domestication traits to wild tomato relatives through modification of key genes involved in day-length sensitivity, shoot architecture, flower and fruit production, and nutrient content (Lemmon et al. 2018; Li et al. 2018d; Zsögön et al. 2018). Potentially, existing wild relatives that already contain superior stress and disease tolerance traits can be edited with CRISPR/Cas tools to have desirable agronomic traits. This de novo domestication maybe faster than conventional breeding strategies for the introgression of disease resistance and stress tolerance into cultivated species from wild relatives due to their polygenic nature (Zsögön et al. 2018). Alternatively, multiplex genome editing may be performed on elite commercial cultivars to engineer and stack different *R* alleles and defense-related genes for broad-spectrum and durable disease resistance.

Because of their ability to specifically cleave DNA or RNA, the CRISPR/Cas system has been utilized for interference and resistance against plant pathogenic RNA and DNA viruses by targeting conserved elements of the viral genome (Ali et al. 2018; Khan et al. 2018; Mushtaq et al. 2020). With the ability to cleave the DNA genome of tomato yellow leaf curl virus (a member of the genus *Geminivirus*), *Nicotiana benthamiana* transgenic plants expressing gRNA/Cas9 were shown to have reduced disease symptoms and lower viral titers (Ali et al. 2015; Baltes et al. 2015; Ji et al. 2015). Furthermore, transient or stable expression of RNA-degrading Cas13a nuclease in *Nicotiana benthamiana* provided viral interference against the RNA viruses, turnip mosaic virus, a member of the genus *Potyvirus* (Aman et al. 2018). Some limitations with this technology include classification of the plants as transgenic, potential nonspecific cleavage by the Cas nuclease, and the evolution of viral resistance to the targeting gRNA. With regards to the latter, Mehta et al. (2019) have found that about 33 to 48% of the transgenic cassava plants expressing Cas9 and a single gRNA failed to confer resistance to African cassava mosaic virus. Viral genomes isolated from susceptible plants revealed point mutations at the viral target site conferring resistance to the CRISPR/Cas9 cleavage (Mehta et al. 2019).

Government regulations and public concerns with genetically modified organism (GMO) are important issues associated with practical application and commercialization of CRISPR/Cas-edited crops in the United States and around the world. The U.S. Department of Agriculture has determined that certain genome-edited crops will not be regulated as GMOs, if the genetic modification involves deletion of any size DNA fragment, substitution of single nucleotide that could otherwise result from chemical or radiation mutagenesis, or insertion of DNA segment that could be achieved via traditional breeding with a sexually compatible species (Wolt and Wolf 2018). On the other hand, the Court of Justice of the European Union (CJEU) ruled that genome-edited crops created by CRISPR/Cas are to be regulated the same as GMOs (Smyth and Lassoued 2018). These contrasting regulatory approaches based on the process versus the product have created a convoluted landscape for politician, scientists, and the general public alike to navigate when it comes to policy, ethics, and commercialization of gene-edited crops. Clear and uniform positions on government regulations among countries globally are needed to maximize the impact of CRISPR technologies.

CRISPR/CAS APPLICATIONS IN PATHOGEN DETECTION AND DISEASE DIAGNOSIS

Beyond its conventional applications in genome editing, CRISPR/Cas has been harnessed for specific and ultra-sensitive detection of DNA or RNA (Fig. 2C). Cas nuclease-based diagnostics is promising technology with potentially broad applications in plant pathogen detection and disease management. Both Cas12a- and Cas13a-based detection harness the specificity of Cas nucleases to detect a defined target sequence in pathogen DNA or RNA (Gootenberg et al. 2017; Li et al. 2018c; Chen et al. 2018). Typically, isothermal amplification in the form of T7-RPA (T7 transcription-recombinase polymerase amplification) (Cas13a specific), RPA, or loop-mediated isothermal amplification (LAMP) is performed prior to the addition of the Cas nuclease. In the cases of SHERLOCK, SHERLOCK v2, HOLMES, and DETECTR methods, Cas nuclease remain activated, allowing for the degradation and quantification of a fluorescent reporter oligonucleotide. In addition, Cas-based methods are compatible with lateral flow assays using immunostrips (Gootenberg et al. 2018). In comparison with traditional nucleic acid detection techniques such as PCR, qPCR, and digital droplet PCR, the Cas-based detection allows significantly lower inputs, with reliable detection at femtomolar (10^{-15}) to zeptomolar sensitivity (10^{-21}) depending upon the methods (Li et al. 2019a). Therefore, CRISPR/Cas-based diagnostics are excellent choices for highly sensitive, specific, rapid, cost-efficient, and multiplex detection of nucleic acids that is capable of supporting point-of-care use (Gootenberg et al. 2018).

The Cas12a-based method has been used for specific and sensitive detection of *M. oryzae*, which is the rice blast fungus and relatively abundant in the infected tissue of rice plants (Zhang et al. 2020; Kang et al. 2020). Cas12a-based visual assays have also been developed for rapid detection of grapevine red-blotch DNA virus as well as apple RNA viruses and viroids (Li et al. 2019b; Jiao et al. 2021). Due to its single nucleotide specificity and extreme sensitivity, Cas-based diagnostics bodes well for early detection of low titer plant pathogens such as '*Candidatus* Liberibacter asiaticus' (Huanglongbing or citrus greening pathogen) and phytoplasmas in the asymptomatic tissue. With portable fluorescence signal detector or immunostrip, the Cas-based diagnostics can pave a way toward rapid and large scale in-field detection of plant pathogens, which will be valuable for qualitative and quantitative measurements of disease prevalence in plants, insect vectors and soils. The field-deployable detection can help establish quarantine zones of invasive pathogens and/or aid in integrated pest management strategies for mitigating the spread of disease, including informed and precise applications of pesticides in the infected field.

CONCLUSION AND FUTURE PERSPECTIVES

CRISPR technologies have had considerable impact on genome engineering of plant disease resistance and will greatly facilitate the basic research of plant−microbe interactions and disease diagnostics in the near future. Although controversial, introduction of CRISPR/Cas-enabled gene drives in insect vectors and plant pathogens may be considered as new strategies for plant disease management (Pixley et al. 2019). Gene drive strategies rely on the Mendelian inheritance of a self-sustaining genetic mutations to spread throughout a population and may allow the genetic control of specific insect or pathogen populations, mitigating the spread and impact of plant disease (Hammond and Galizi 2017). Gene drives risk of the development of adaptive genetic resistance. However, a number of strategies have been developed to mitigate potential resistance in future applications (Hammond and Galizi 2017). Plant genome editing is limited by the delivery of CRISPR/Cas machinery and the requirement of tissue culture for plant regeneration. Recent developments point toward novel techniques exploiting nanomaterials or viral vectors for CRISPR/Cas delivery and in planta genome editing (Cunningham et al. 2018; Ellison et al. 2020). Despite the potential and promises for practical applications in plant pathology and disease management, various technical, regulatory and social limitations associated with the commercialization of genome editing technologies must be properly resolved for the world agricultural industry to benefit from the CRISPR revolution.

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